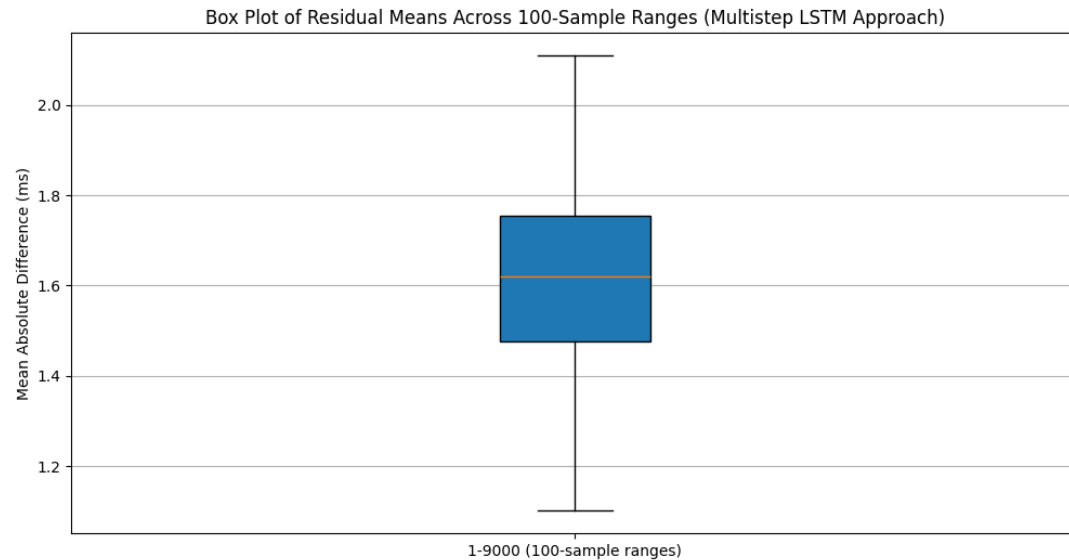


BOX PLOT OF RESIDUAL MEANS

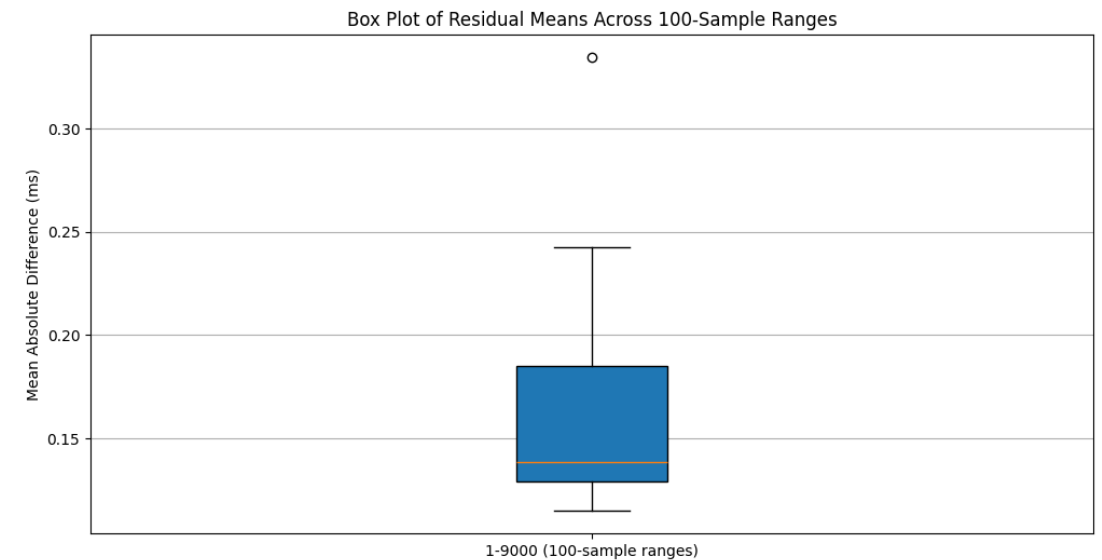
For Normal Multistep LSTM Approach



Residual Means Summary:

	Range_Start	Range_End	Residual_Mean
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	1.610396
std	2612.470096	2612.470096	0.239306
min	1.000000	100.000000	1.100857
25%	2226.000000	2325.000000	1.476537
50%	4451.000000	4550.000000	1.621205
75%	6676.000000	6775.000000	1.754922
max	8901.000000	9000.000000	2.109273

For Hybrid LSTM Approach (Block Size = 10)

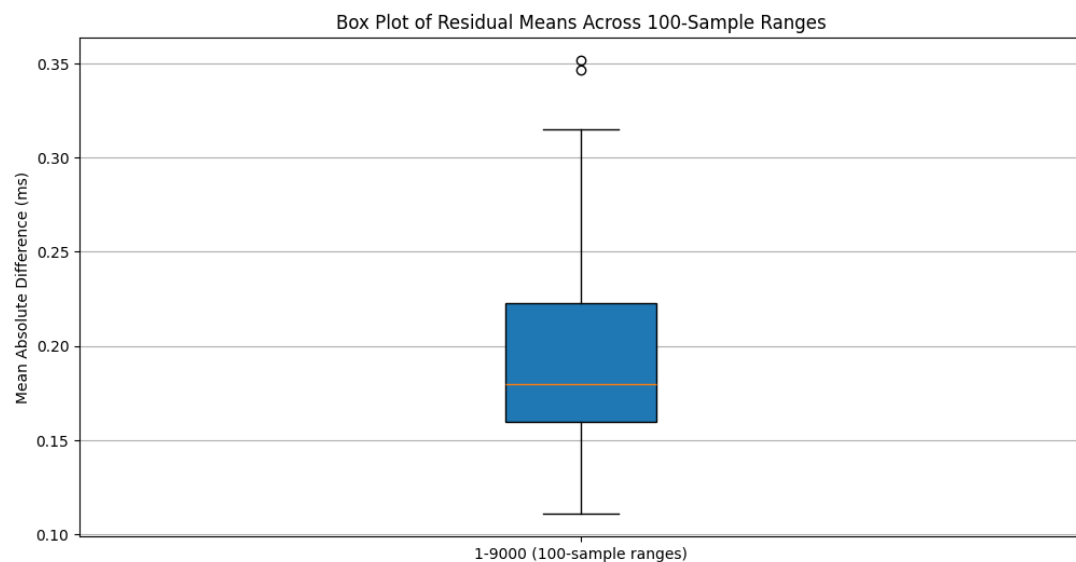


Residual Means Summary:

	Range_Start	Range_End	Residual_Mean
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.156315
std	2612.470096	2612.470096	0.038674
min	1.000000	100.000000	0.114883
25%	2226.000000	2325.000000	0.129258
50%	4451.000000	4550.000000	0.138333
75%	6676.000000	6775.000000	0.184905
max	8901.000000	9000.000000	0.334327

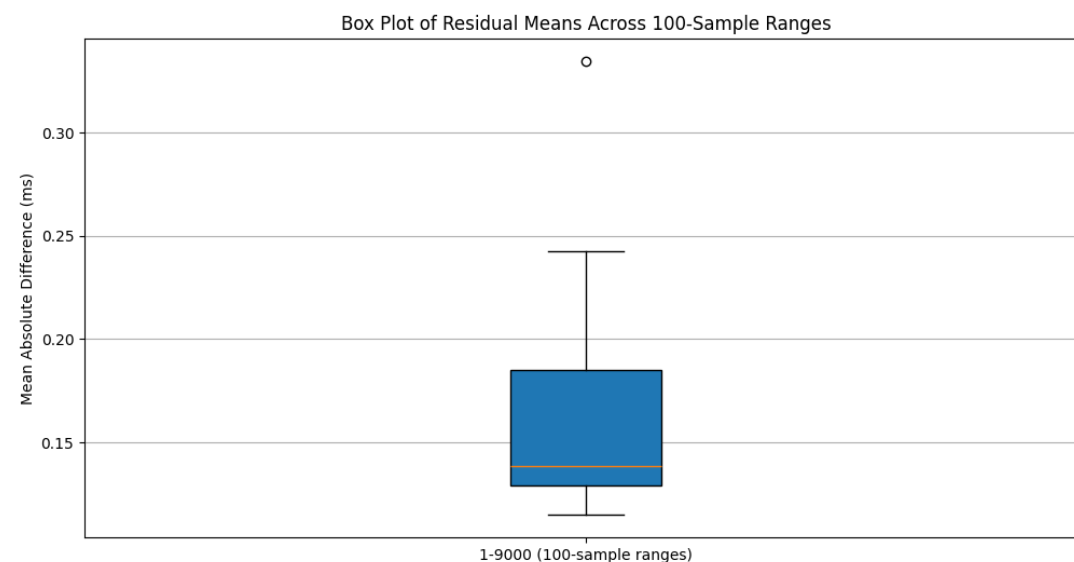
BOX PLOT OF RESIDUAL MEANS

- For Hybrid LSTM Approach (Block Size = 30)
- For Hybrid LSTM Approach (Block Size = 10)



Residual Means Summary:

	Range_Start	Range_End	Residual_Mean
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.196134
std	2612.470096	2612.470096	0.050204
min	1.000000	100.000000	0.110839
25%	2226.000000	2325.000000	0.159720
50%	4451.000000	4550.000000	0.180032
75%	6676.000000	6775.000000	0.222504
max	8901.000000	9000.000000	0.351552

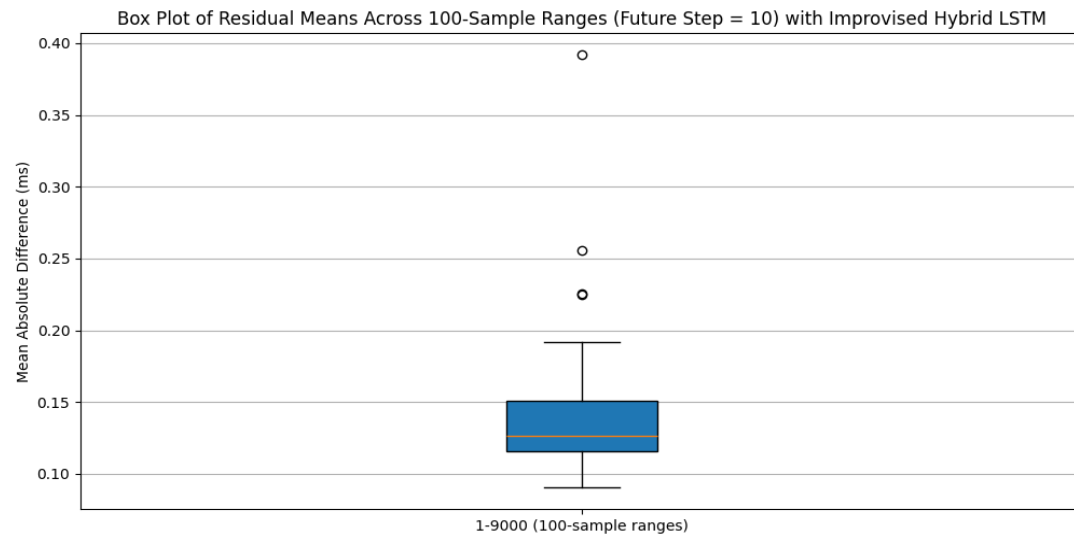


Residual Means Summary:

	Range_Start	Range_End	Residual_Mean
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.156315
std	2612.470096	2612.470096	0.038674
min	1.000000	100.000000	0.114883
25%	2226.000000	2325.000000	0.129258
50%	4451.000000	4550.000000	0.138333
75%	6676.000000	6775.000000	0.184905
max	8901.000000	9000.000000	0.334327

BOX PLOT OF RESIDUAL MEANS

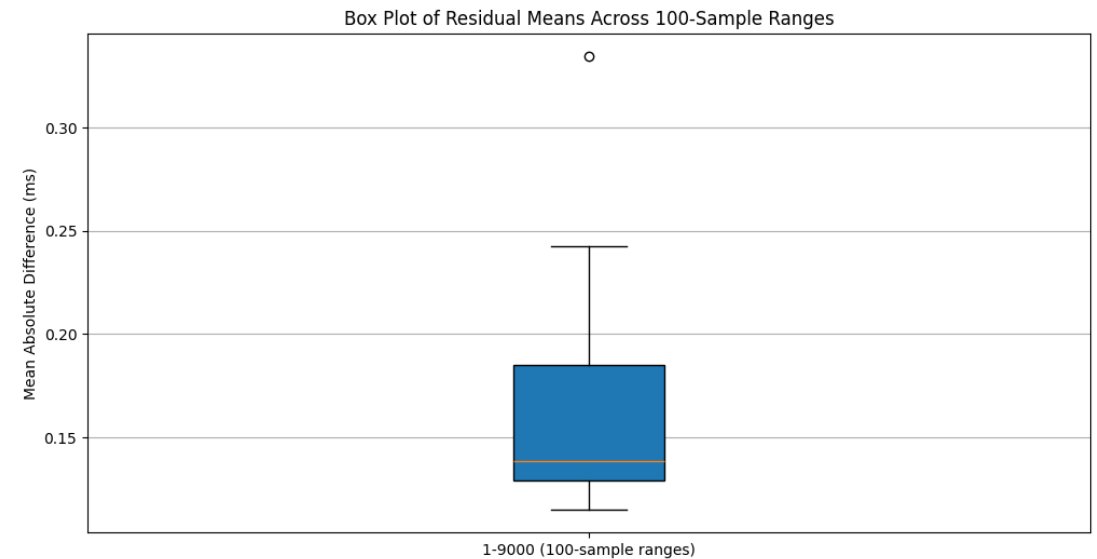
- For Improvised Hybrid LSTM Approach (Block Size = 10)



Residual Means Summary:

	Range_Start	Range_End	Residual_Mean
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.138724
std	2612.470096	2612.470096	0.041134
min	1.000000	100.000000	0.090322
25%	2226.000000	2325.000000	0.115678
50%	4451.000000	4550.000000	0.126485
75%	6676.000000	6775.000000	0.150647
max	8901.000000	9000.000000	0.391850

- For Hybrid LSTM Approach (Block Size = 10)

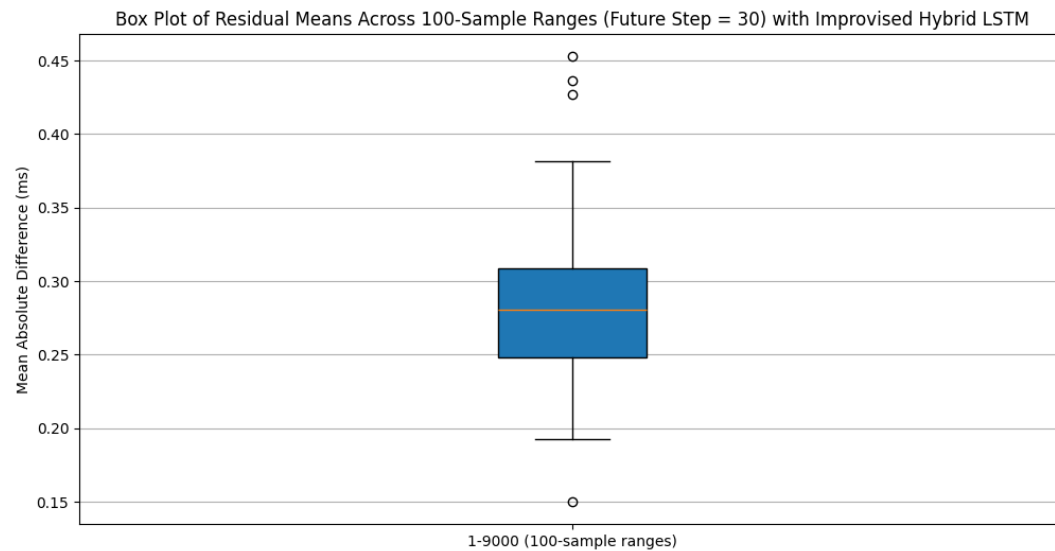


Residual Means Summary:

	Range_Start	Range_End	Residual_Mean
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.156315
std	2612.470096	2612.470096	0.038674
min	1.000000	100.000000	0.114883
25%	2226.000000	2325.000000	0.129258
50%	4451.000000	4550.000000	0.138333
75%	6676.000000	6775.000000	0.184905
max	8901.000000	9000.000000	0.334327

BOX PLOT OF RESIDUAL MEANS

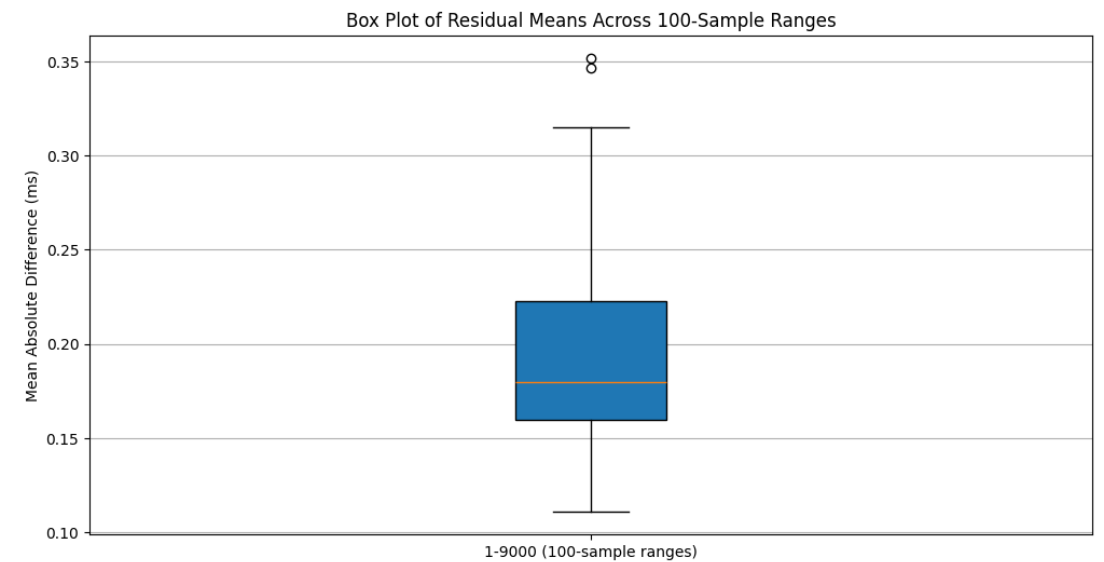
- For Improvised Hybrid LSTM Approach (Block Size = 30)



Residual Means Summary:

	Range_Start	Range_End	Residual_Mean
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.283511
std	2612.470096	2612.470096	0.052159
min	1.000000	100.000000	0.149863
25%	2226.000000	2325.000000	0.248070
50%	4451.000000	4550.000000	0.280518
75%	6676.000000	6775.000000	0.309054
max	8901.000000	9000.000000	0.452738

- For Hybrid LSTM Approach (Block Size = 30)

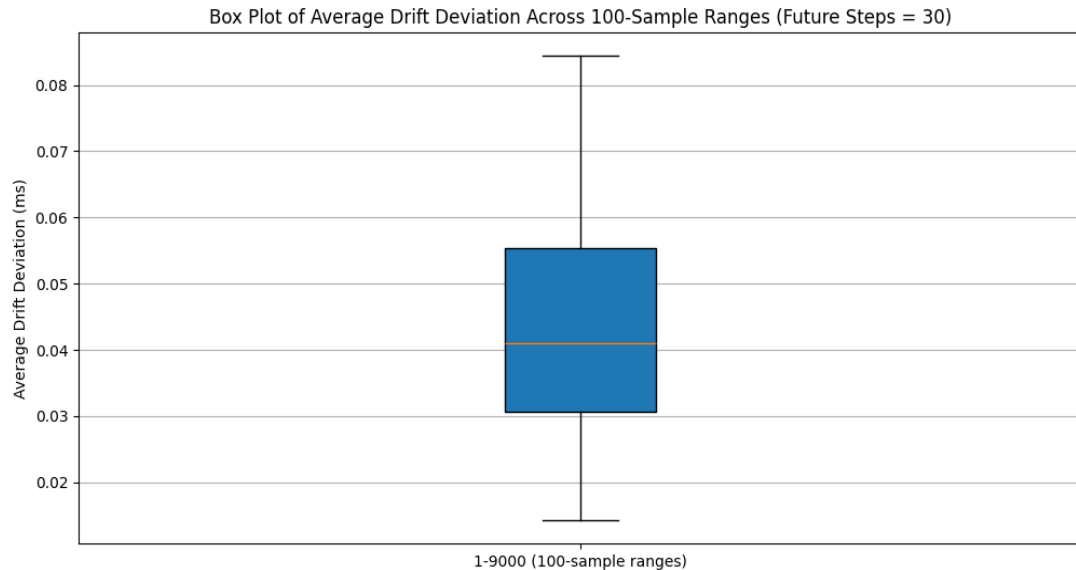


Residual Means Summary:

	Range_Start	Range_End	Residual_Mean
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.196134
std	2612.470096	2612.470096	0.050204
min	1.000000	100.000000	0.110839
25%	2226.000000	2325.000000	0.159720
50%	4451.000000	4550.000000	0.180032
75%	6676.000000	6775.000000	0.222504
max	8901.000000	9000.000000	0.351552

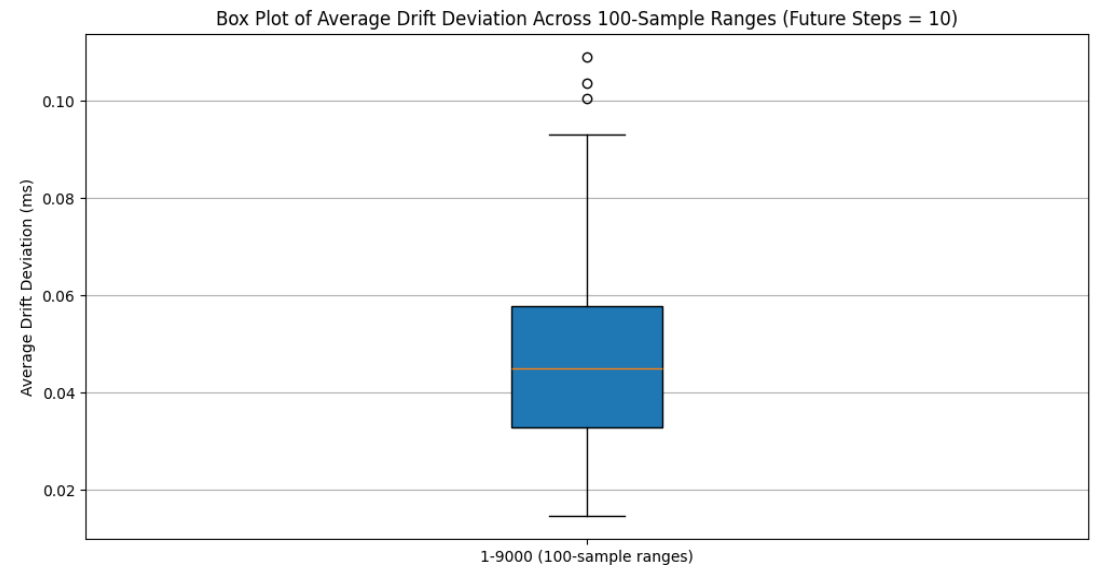
BOX PLOT OF AVERAGE DRIFT DEVIATION

- For Hybrid LSTM Approach (Block Size = 30)
- For Hybrid LSTM Approach (Block Size = 10)



Average Drift Deviation Summary:

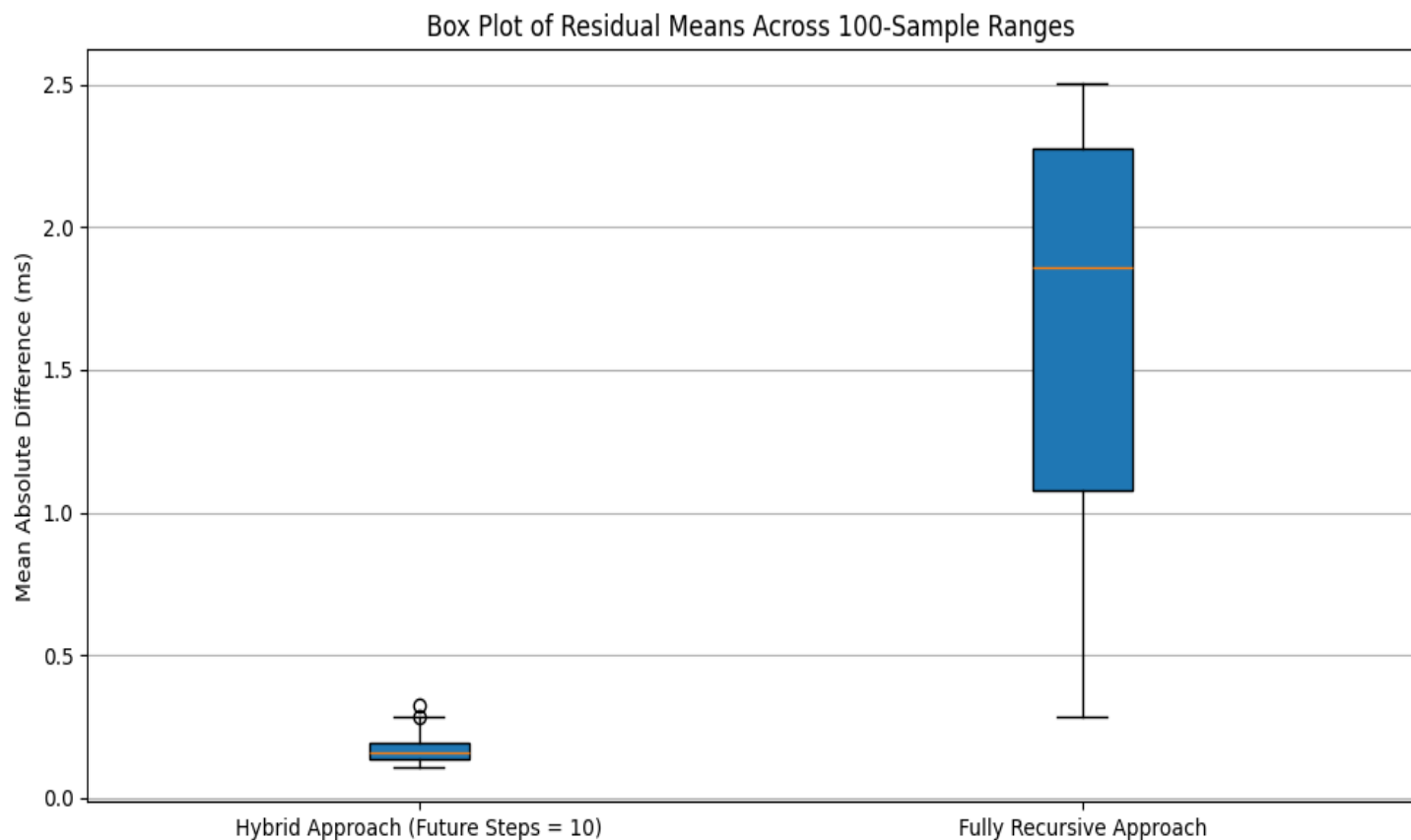
	Range_Start	Range_End	Average_Drift_Deviation
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.043894
std	2612.470096	2612.470096	0.017850
min	1.000000	100.000000	0.014226
25%	2226.000000	2325.000000	0.030594
50%	4451.000000	4550.000000	0.040952
75%	6676.000000	6775.000000	0.055334
max	8901.000000	9000.000000	0.084406



Average Drift Deviation Summary:

	Range_Start	Range_End	Average_Drift_Deviation
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.046875
std	2612.470096	2612.470096	0.021658
min	1.000000	100.000000	0.014635
25%	2226.000000	2325.000000	0.032979
50%	4451.000000	4550.000000	0.045025
75%	6676.000000	6775.000000	0.057722
max	8901.000000	9000.000000	0.108916

BOX PLOT OF RESIDUAL MEANS FOR LSTM WITH HYBRID APPROACH FOR FS=10 & FS=30



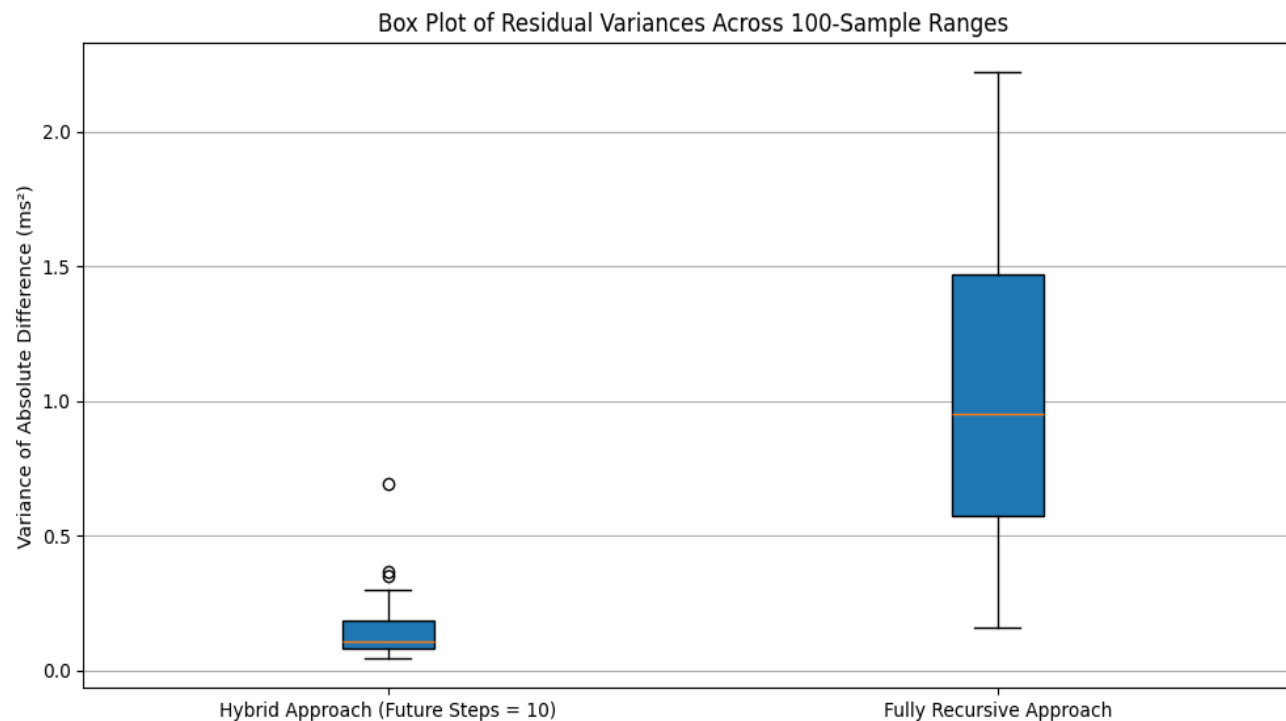
Residual Means Summary (Hybrid):

	Range_Start	Range_End	Residual_Mean
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.168662
std	2612.470096	2612.470096	0.043296
min	1.000000	100.000000	0.105668
25%	2226.000000	2325.000000	0.135829
50%	4451.000000	4550.000000	0.161161
75%	6676.000000	6775.000000	0.194682
max	8901.000000	9000.000000	0.322531

Residual Means Summary (Fully Recursive):

	Range_Start	Range_End	Residual_Mean
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	1.622690
std	2612.470096	2612.470096	0.719536
min	1.000000	100.000000	0.283834
25%	2226.000000	2325.000000	1.075089
50%	4451.000000	4550.000000	1.859042
75%	6676.000000	6775.000000	2.275775
max	8901.000000	9000.000000	2.502367

BOX PLOT OF VARIANCE FOR HYBRID AND FULLY RECURSIVE APPROACH



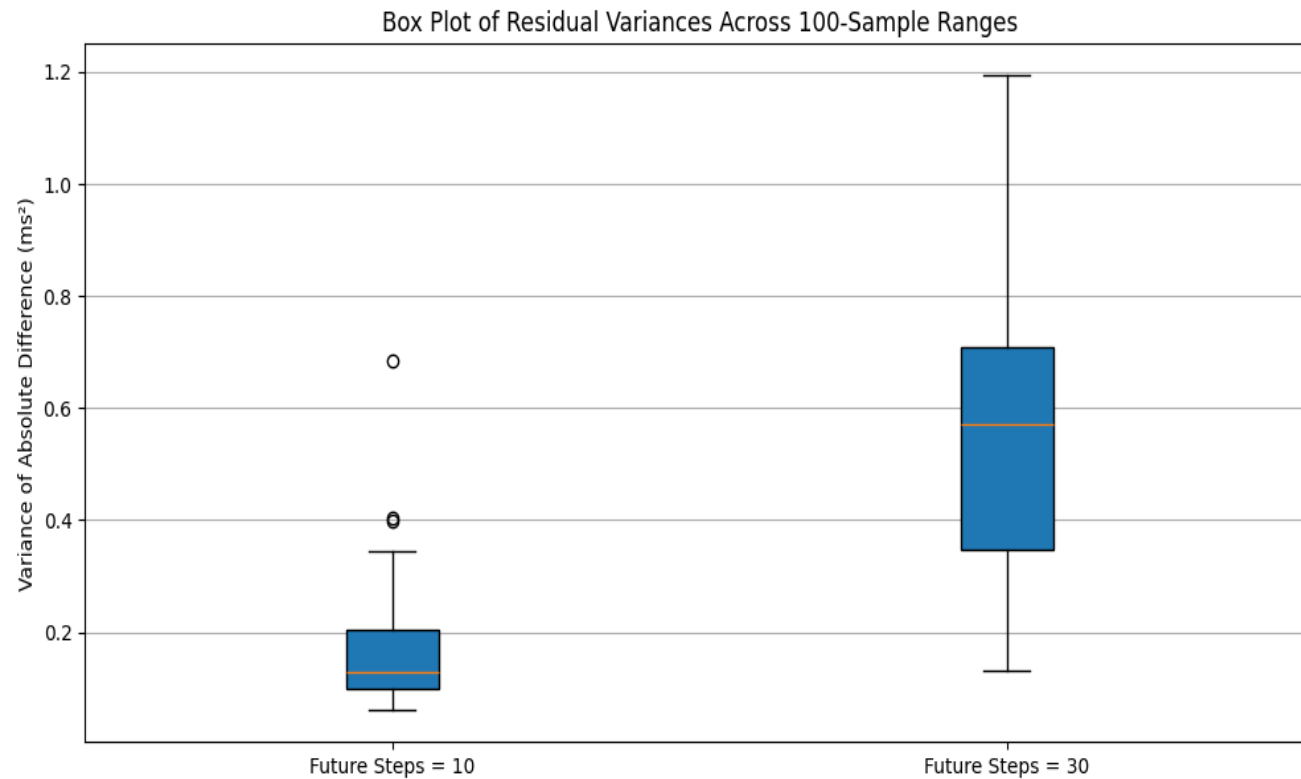
Residual Variances Summary (Hybrid):

	Range_Start	Range_End	Residual_Variance
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.139363
std	2612.470096	2612.470096	0.095594
min	1.000000	100.000000	0.043039
25%	2226.000000	2325.000000	0.078741
50%	4451.000000	4550.000000	0.109103
75%	6676.000000	6775.000000	0.182682
max	8901.000000	9000.000000	0.690697

Residual Variances Summary (Fully Recursive):

	Range_Start	Range_End	Residual_Variance
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	1.037379
std	2612.470096	2612.470096	0.527384
min	1.000000	100.000000	0.160009
25%	2226.000000	2325.000000	0.573473
50%	4451.000000	4550.000000	0.953757
75%	6676.000000	6775.000000	1.469720
max	8901.000000	9000.000000	2.219675

BOX PLOT OF VARIANCE FOR LSTM WITH HYBRID APPROACH FOR FS=10 & FS=30



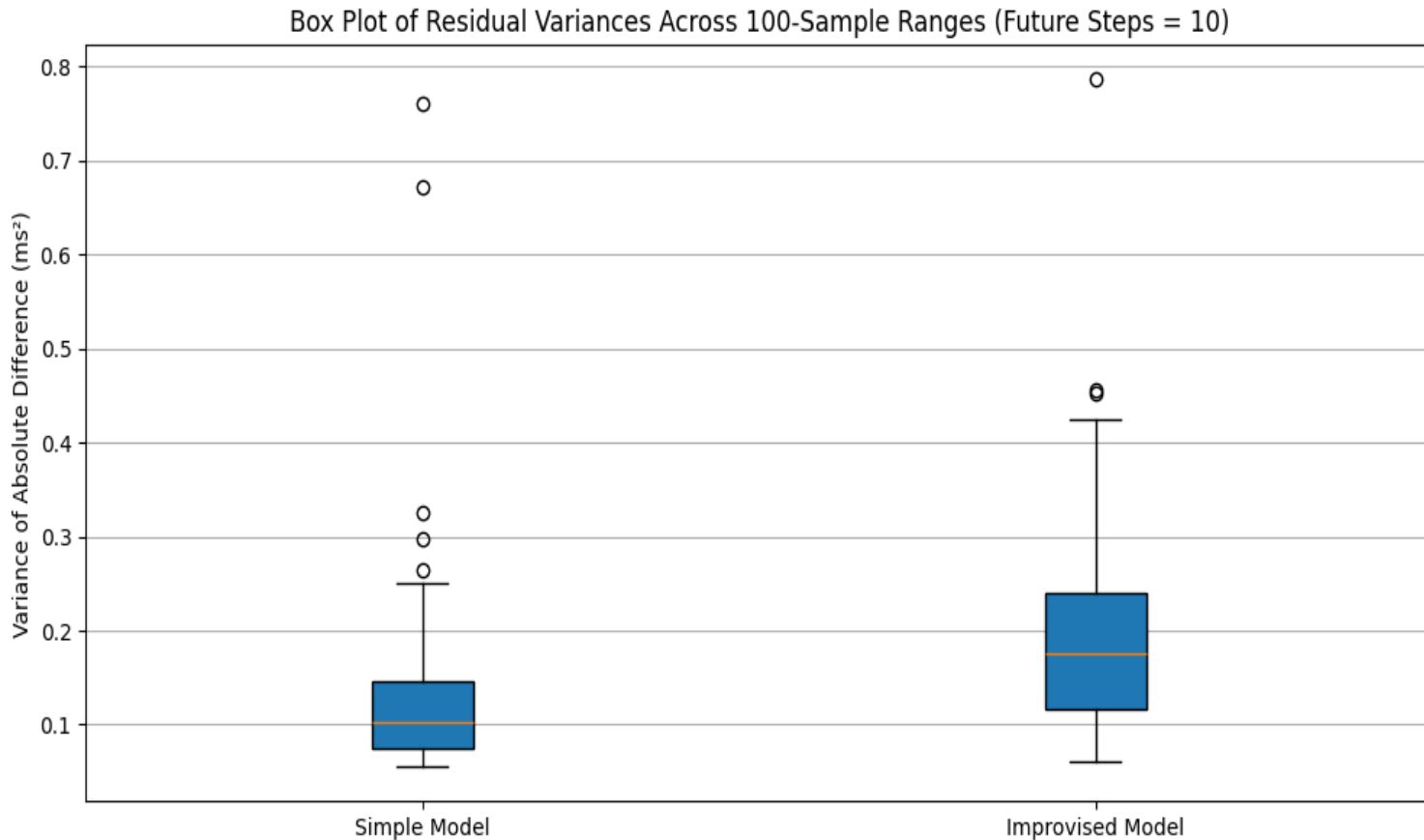
Residual Variances Summary (Future Steps = 10):

	Range_Start	Range_End	Residual_Variance
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.165766
std	2612.470096	2612.470096	0.100254
min	1.000000	100.000000	0.060575
25%	2226.000000	2325.000000	0.099001
50%	4451.000000	4550.000000	0.128200
75%	6676.000000	6775.000000	0.204444
max	8901.000000	9000.000000	0.685429

Residual Variances Summary (Future Steps = 30):

	Range_Start	Range_End	Residual_Variance
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.555636
std	2612.470096	2612.470096	0.246697
min	1.000000	100.000000	0.130384
25%	2226.000000	2325.000000	0.347521
50%	4451.000000	4550.000000	0.570640
75%	6676.000000	6775.000000	0.708328
max	8901.000000	9000.000000	1.193700

BOX PLOT OF VARIANCE FOR LSTM WITH HYBRID AND IMPROVISED HYBRID APPROACH WHEN FS=10



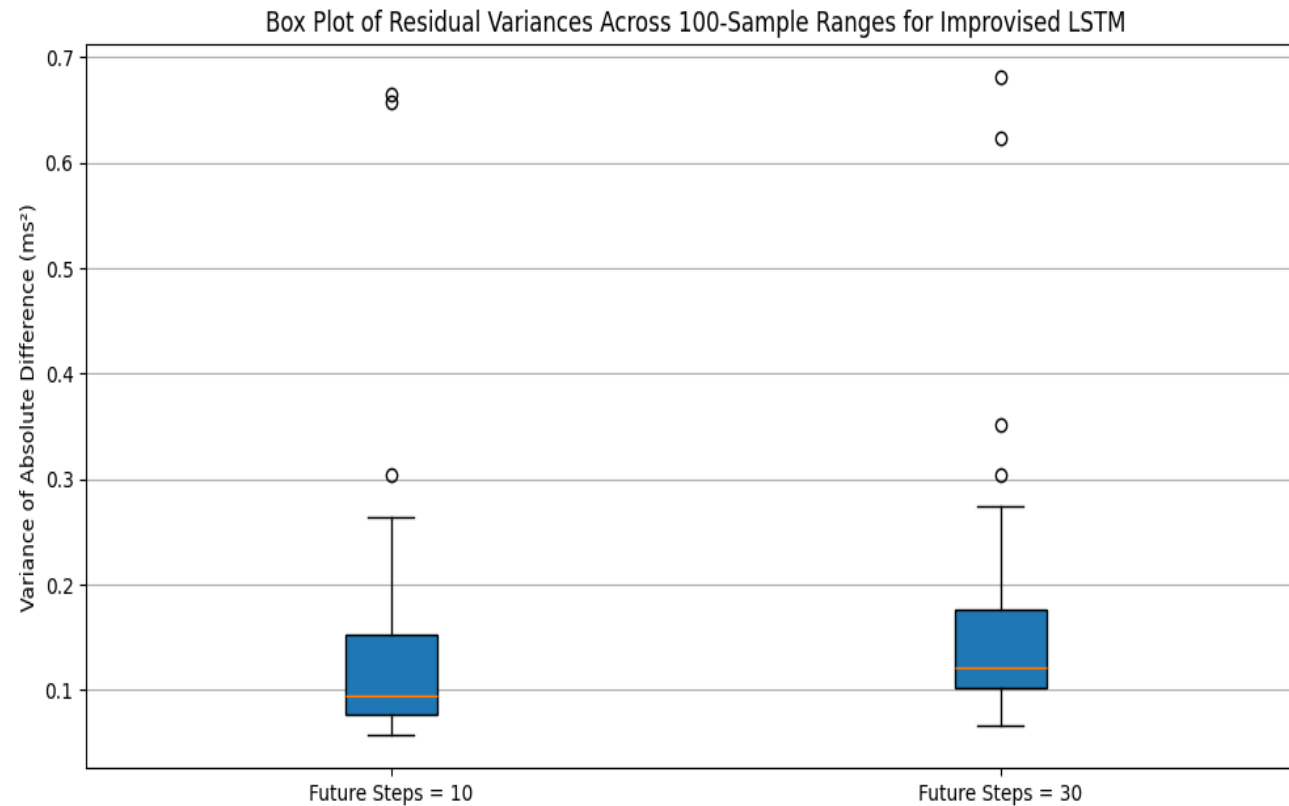
Residual Variances Summary (Simple Model):

	Range_Start	Range_End	Residual_Variance
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.133888
std	2612.470096	2612.470096	0.106261
min	1.000000	100.000000	0.055146
25%	2226.000000	2325.000000	0.074309
50%	4451.000000	4550.000000	0.103205
75%	6676.000000	6775.000000	0.146480
max	8901.000000	9000.000000	0.760843

Residual Variances Summary (Improved Model):

	Range_Start	Range_End	Residual_Variance
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.198536
std	2612.470096	2612.470096	0.120949
min	1.000000	100.000000	0.060186
25%	2226.000000	2325.000000	0.117032
50%	4451.000000	4550.000000	0.175035
75%	6676.000000	6775.000000	0.240770
max	8901.000000	9000.000000	0.785934

BOX PLOT OF VARIANCE FOR LSTM WITH IMPROVISED HYBRID APPROACH FOR FS=10 & FS=30



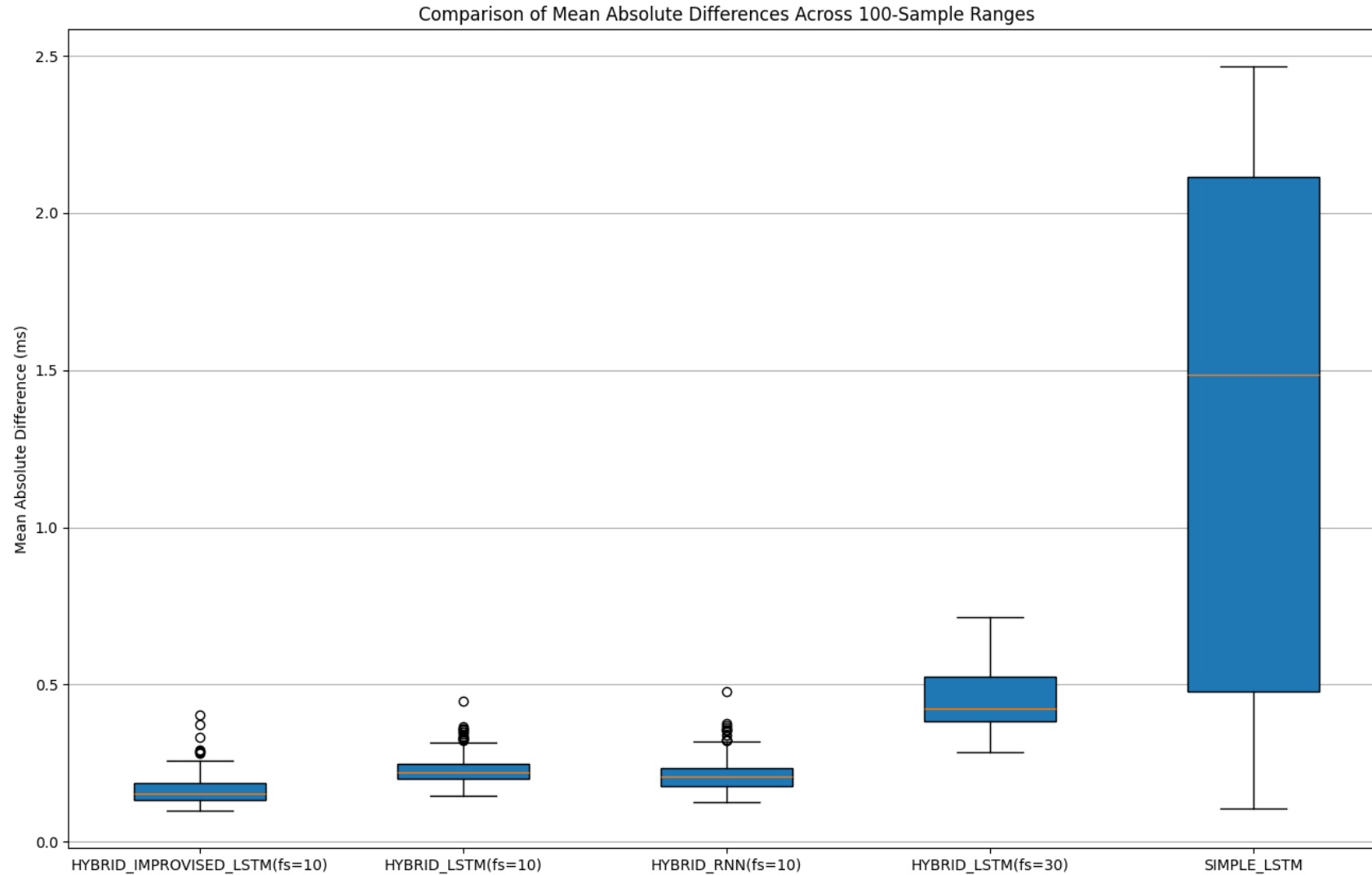
Residual Variances Summary (Future Steps = 10):

	Range_Start	Range_End	Residual_Variance
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.126194
std	2612.470096	2612.470096	0.095283
min	1.000000	100.000000	0.056995
25%	2226.000000	2325.000000	0.077211
50%	4451.000000	4550.000000	0.094145
75%	6676.000000	6775.000000	0.152862
max	8901.000000	9000.000000	0.664074

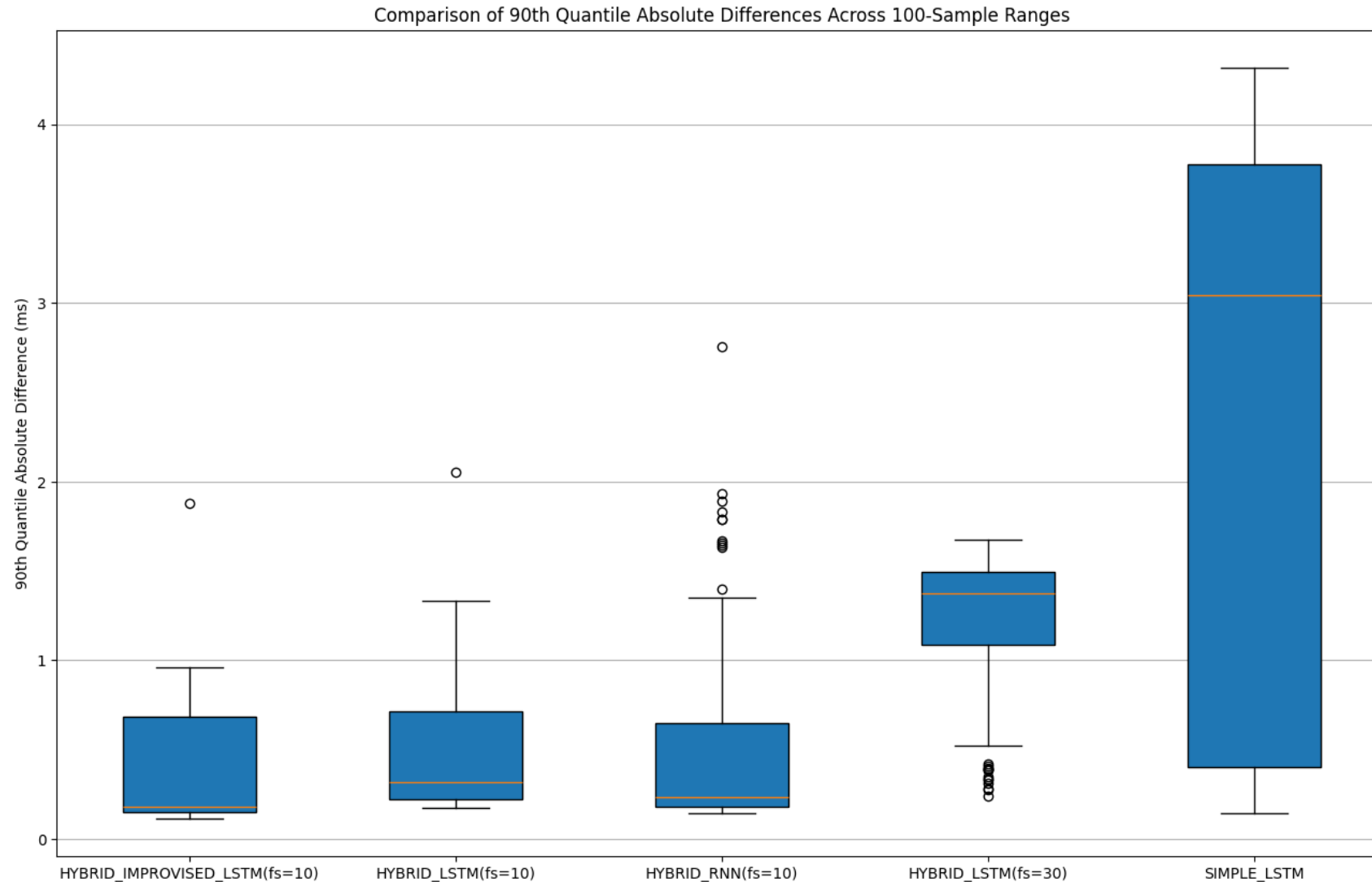
Residual Variances Summary (Future Steps = 30):

	Range_Start	Range_End	Residual_Variance
count	90.000000	90.000000	90.000000
mean	4451.000000	4550.000000	0.152198
std	2612.470096	2612.470096	0.093290
min	1.000000	100.000000	0.066967
25%	2226.000000	2325.000000	0.102293
50%	4451.000000	4550.000000	0.121723
75%	6676.000000	6775.000000	0.176285
max	8901.000000	9000.000000	0.680736

Comparison of Mean Absolute Differences Across 100-Sample Ranges



Comparison of 90th Quantile Absolute Differences Across 100-Sample Ranges



Comparison of Variance of Absolute Differences Across 100-Sample Ranges

